

1. Static Games of Complete Information

Normal-Form Game Representation

The normal-form (strategic-form) representation of an n -player game specifies the strategy spaces S_i and payoff functions u_i , denoted as $G = \{S_1, \dots, S_n; u_1, \dots, u_n\}$

Dominant Strategies

In game G , strategy s'_i *strictly dominates* strategy s''_i if:

$$u_i(s'_i, s_{-i}) > u_i(s''_i, s_{-i}), \quad \forall s_{-i} \in S_{-i}$$

Nash Equilibrium

Best Response : The best response for player i to strategies s_{-i} is:

$$R_i(s_{-i}) = \arg \max_{s_i \in S_i} u_i(s_i, s_{-i})$$

Nash equilibrium : Strategies $(s_1^*, \dots, s_n^*)$ form a Nash equilibrium if:

$$s_i^* \in R_i(s_{-i}^*), \quad \forall i = 1, \dots, n$$

Equivalently: $u_i(s_i^*, s_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, s_{-i}^*)$

Mixed Strategies

A mixed strategy for player i is a probability distribution $p_i = (p_{i1}, \dots, p_{iK})$ over pure strategies, where $\sum_k p_{ik} = 1$ and $p_{ik} \geq 0$

Expected Payoff : For a 2-player game:

$$v_1(p_1, p_2) = \sum_{j=1}^J \sum_{k=1}^K p_{1j} p_{2k} u_1(s_{1j}, s_{2k})$$
$$v_2(p_1, p_2) = \sum_{j=1}^J \sum_{k=1}^K p_{1j} p_{2k} u_2(s_{1j}, s_{2k})$$

Mixed strategies (p_1^*, p_2^*) form a Nash equilibrium if:

$$v_1(p_1^*, p_2^*) \geq v_1(p_1, p_2^*)$$

$$v_2(p_1^*, p_2^*) \geq v_2(p_1^*, p_2)$$

for all probability distributions p_1 and p_2

Theorem (Nash (1950)): In any finite n -player normal-form game, there exists at least one Nash equilibrium, possibly involving mixed strategies.

2. Dynamic Games of Complete Information

Dynamic Games

A dynamic game of complete and perfect information is a game where:

- Players move in sequence
- Perfect information: all previous moves are observed before next move is chosen
- Complete information: payoffs are common knowledge

Such games can be represented by a game tree.

Backwards Induction

- At the second stage, player 2 observes a_1 and solves $\max_{a_2 \in A_2} u_2(a_1, a_2)$
- The solution is player 2's best response $R_2(a_1)$
- Knowing this, player 1 solves $\max_{a_1 \in A_1} u_1(a_1, R_2(a_1))$
- The solution a_1^* and $R_2(a_1^*)$ is the backwards-induction outcome

Extensive-Form Games and Strategies

The extensive-form representation of a game specifies:

- The players in the game
- When each player has the move, what each player can do, and what each player knows when moving
- The payoffs received for each combination of moves

Information Set : A collection of decision nodes satisfying:

- The player needs to move at every node in the set
- When play reaches a node in the set, the player does not know which node in the set has been reached
- The player must have the same actions available at each node in the set

Strategy : A complete plan of action specifying a feasible action for the player in every contingency where the player might be called on to act.

Subgame-Perfect Nash Equilibrium

Subgame : A portion of a game that:

- Begins at a decision node that is a singleton information set
- Includes all nodes following it in the game tree
- Does not cut any information sets

A Nash equilibrium is **subgame-perfect** if the players' strategies constitute a Nash equilibrium in every subgame.

3. Static Games of Incomplete Information

Bayesian Games Games of incomplete information are also called Bayesian games. In a game of incomplete information, at least one player is uncertain about another player's payoff function.

Static Bayesian Games and Bayesian Nash Equilibrium

The normal-form representation of an n -player static Bayesian game specifies:

- The players' action spaces $A_1, \dots, A_n$,
- Their type spaces $T_1, \dots, T_n$,
- Their beliefs $P_1, \dots, P_n$, and
- Their payoff functions $u_1, \dots, u_n$.

- Moreover, Player i 's type, $t_i \in T_i$, is privately known by player i , and determines player i 's payoff function, $u_i(a_1, \dots, a_n; t_i)$.
- Player i 's belief $P_i(t_{-i}|t_i)$ describes i 's uncertainty about the $n - 1$ other players' possible types, t_{-i} , given i 's own type, t_i .

Strategy : In the static Bayesian game, a strategy for player i is a function $s_i : T_i \rightarrow A_i$.

- For given type t_i , $s_i(t_i)$ gives the action.
- Player i 's strategy space S_i is the set of all functions from T_i into A_i .

In the static Bayesian game, the strategies $s^* = (s_1^*, \dots, s_n^*)$ are a (pure-strategy) **Bayesian Nash equilibrium** if for each player i and for each of i 's type t_i in T_i , $s_i^*(t_i)$ solves

$$\max_{a_i \in A_i} E_{t_{-i}} u_i(a_i, s_{-i}^*(t_{-i}); t_i),$$

where

$$E_{t_{-i}} u_i(a_i, s_{-i}^*(t_{-i}); t_i) = \sum_{t_{-i} \in T_{-i}} P_i(t_{-i}|t_i) u_i(a_i, s_{-i}^*(t_{-i}); t_i).$$

Theorem (Proposition 1): Let $(s_1^*, \dots, s_n^*)$ be strategies in a static Bayesian game. If for any $t_i \in T_i$, $a_i \in A_i$ and $a_{-i} \in A_{-i}$,

$$u_i(s_i^*(t_i), a_{-i}; t_i) \geq u_i(a_i, a_{-i}; t_i),$$

(i.e., $s_i^*(t_i)$ weakly dominates every $a_i \in A_i$), then $(s_1^*, \dots, s_n^*)$ is a Bayesian Nash equilibrium.

4. Dynamic Games of Incomplete Info

Perfect Bayesian Equilibrium

A **Perfect Bayesian Equilibrium** consists of strategies and beliefs satisfying:

1. **Sequential Rationality:** Given their beliefs, players' strategies must be sequentially rational.
 - At each information set, the action and subsequent strategy of the player must be optimal, given the player's belief and other players' subsequent strategies.
2. **Consistency:** At every information set, beliefs are determined by Bayes' rule and players' strategies, whenever possible.

Remark: The strategy profile in any perfect Bayesian equilibrium is a Nash equilibrium.

Signaling Games

Pooling vs. Separating:

- **Pooling:** Each type sends the same message
- **Separating:** Each type sends a different message
- **Semi-separating/Partial pooling**

A **pure-strategy perfect Bayesian equilibrium** in a signaling game consisting of Sender's strategy $m^*(t_i)$, Receiver's strategy $a^*(m_j)$, Belief $\mu(t_i|m_j)$ satisfying:

1. Sequential rationality:

- For each $m_j \in M$, $a^*(m_j)$ maximizes the Receiver's expected utility:

$$a^*(m_j) = \arg \max_{a_k \in A} \sum_{t_i \in T} \mu(t_i|m_j) U_R(t_i, m_j, a_k)$$

- For each $t_i \in T$, $m^*(t_i)$ maximizes the Sender's utility:

$$m^*(t_i) = \arg \max_{m_j \in M} U_S(t_i, m_j, a^*(m_j))$$

2. **Consistency:** Given Sender's strategy m^* , for each m_j with $T_j = \{t_i \in T : m^*(t_i) = m_j\} \neq \emptyset$:

$$\mu(t_i|m_j) = \frac{P(t_i)}{\sum_{t \in T_j} P(t)} \text{ for all } t_i \in T_j$$

5. Cooperative Games

Two-Person Cooperative Games

A pair $\Gamma = (H, d)$ where:

- $H \subset \mathbb{R}^2$ is compact and convex
- $d \in H$ is the threat point (payoffs if no agreement)
- H contains at least one element, u , such that $u \gg d$

Pareto optimality : Payoff pairs which are not dominated by any other pair are said to be Pareto optimal. Pair (u, v) dominates (u', v') if $u \geq u'$ and $v \geq v'$.

Nash bargaining solution : A mapping $f : W \rightarrow \mathbb{R}^2$ that associates a unique element $f(H, d) = (f_1(H, d), f_2(H, d))$ with the game $(H, d) \in W$, satisfying:

1. **Feasibility:** $f(H, d) \in H$
2. **Individual rationality:** $f(H, d) \geq d$
3. **Pareto optimality:** $f(H, d)$ is Pareto optimal
4. **Invariance under linear transformations:** If a game is transformed by affine functions, the solution transforms accordingly
5. **Symmetry:** If (H, d) is symmetric, then $f_1(H, d) = f_2(H, d)$
6. **Independence of irrelevant alternatives:** If $(H, d), (H', d') \in W$, $d = d'$, $H \subset H'$, and $f(H', d') \in H$, then $f(H, d) = f(H', d')$

Theorem (Theorem 1): A game $(H, d) \in W$ has a unique Nash solution $u^* = f(H, d)$ satisfying the above conditions if and only if $(u_1^* - d_1)(u_2^* - d_2) > (u_1 - d_1)(u_2 - d_2)$ for all $u \in H$, $u \geq d$, and $u \neq u^*$.

n-person Cooperative Games

For an n -person game with players $N = \{1, 2, \dots, n\}$, the **characteristic function** v gives each coalition $S \subseteq N$ the amount $v(S)$ that coalition S can be sure of receiving. Game denoted as $\Gamma = (N, v)$.

Assumption : The characteristic function v satisfies:

- $v(\emptyset) = 0$
- For any disjoint coalitions K and L contained in N , $v(K \cup L) \geq v(K) + v(L)$ (super-additivity)

Imputation : A payoff vector $x = (x_1, \dots, x_n)$ satisfying:

- $\sum_{i=1}^n x_i = v(N)$ (group rationality)
- $x_i \geq v(\{i\})$ for all $i \in N$ (individual rationality)

$I(N, v)$ denotes the set of all imputations.

Domination : Let $x, y \in I(N, v)$, and S be a coalition. We say x dominates y via S (notation $x \succ_S y$) if:

- $x_i > y_i$ for all $i \in S$
- $\sum_{i \in S} x_i \leq v(S)$

We say x dominates y (notation $x \succ y$) if there is any coalition S such that $x \succ_S y$.

Core : The set of all undominated imputations for a game (N, v) is called the core, denoted by $C(N, v)$.

Theorem (Theorem 2): The core of a game (N, v) is the set of all n -vectors, x , satisfying:

- $\sum_{i \in S} x_i \geq v(S)$ for all $\emptyset \neq S \subset N$
- $\sum_{i \in N} x_i = v(N)$

Special Games

Constant-sum game : A game (N, v) is constant-sum if for all $S \subset N$, $v(S) + v(N - S) = v(N)$.

Essential game : A game (N, v) is essential if $v(N) > \sum_{i \in N} v(\{i\})$. It is inessential otherwise.

Theorem (Theorem 3): If (N, v) is an inessential game, then for any coalition S , $v(S) = \sum_{i \in S} v(\{i\})$.

Theorem (Theorem 4): If a game (N, v) is constant-sum, then its core is either empty or a singleton $\{v(\{1\}), \dots, v(\{n\})\}$.

Strategical Equivalence

Strategical equivalence : Two games v and u are strategically equivalent if there exist constants $a > 0$ and $c_1, \dots, c_n$, such that for every coalition S , $u(S) = av(S) + \sum_{i \in S} c_i$.

Reduced form : A characteristic function v is in $(0, 1)$ -reduced form if $v(\{i\}) = 0$ for all $i \in N$ and $v(N) = 1$.

The Shapley Value

Shapley value : Given any n -person game (N, v) , the Shapley value is an n -vector, denoted by $\phi(v)$, where the i -th component is:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{s!(n-s-1)!}{n!} [v(S \cup \{i\}) - v(S)]$$

where $s = |S|$.

Properties of Shapley value : The Shapley value has the following properties:

- **Individual rationality:** $\phi_i(v) \geq v(\{i\})$ for every $i \in N$
- **Efficiency:** $\sum_{i \in N} \phi_i(v) = v(N)$
- **Symmetry:** If $v(S \cup \{i\}) = v(S \cup \{j\})$ for every coalition S not containing i and j , then $\phi_i(v) = \phi_j(v)$
- **Additivity:** If v and w are characteristic functions, then $\phi(v + w) = \phi(v) + \phi(w)$
- **Dummy:** If a player i is such that $v(S \cup \{i\}) = v(S)$ for every S not containing i , then $\phi_i(v) = 0$